# Sets and Frozen Sets

Topics:

- Understanding Sets & Frozen Sets
- Associated Methods

sets

## Sets

Sets are unordered collection of unique elements.

## Creating Sets

sets in Python can be created by just placing the sequence inside the curly brackets { }.

# Examples

```
>>> s = {20,10,20,10,40,30}
>>> print(s)
{40, 10, 20, 30}
>>> print(type(s))
<class 'set'>
```

# Set Methods

Set Object is defined with the following pre-defined Methods

**.add(...)**

Add an element to a set.

This has no effect if the element is already present.

**.clear(...)**

Remove all elements from this set.

**.copy()**

It creates a shallow copy of the Set object

# Set Methods

**.difference(...)**

Return the difference of two or more sets as a new set.

**.difference_update(...)**

Remove all elements of another set from this set.

**.discard(...)**

Remove an element from a set if it is a member.

If the element is not a member, do nothing.

# Set Methods

**.intersection(...)**

Return the intersection of two sets as a new set.

**.intersection_update(...)**

Update a set with the intersection of itself and another.

**.isdisjoint(...)**

Return True if two sets have a null intersection.

**.issubset(...)**

| Report whether another set contains this set.

# Set Methods

**.issuperset(...)**

Report whether this set contains another set.

**.pop(...)**

Remove and return an arbitrary set element.

Raises KeyError if the set is empty.

**.remove(...)**

Remove an element from a set; it must be a member.

If the element is not a member, raise a KeyError.

# Set Methods

**.union(...)**

Return the union of sets as a new set.

(i.e., all elements that are in either set.)

**.update(...)**

Update a set with the union of itself and others.

# Set Examples

```
>>> s = {10,20,10,30,20,40,30,10}
>>> print(s)
{40, 10, 20, 30}
>>> s.add(50)
>>> print(s)
{40, 10, 50, 20, 30}
>>> s.clear()
>>> print(s)
set()
>>> s1 = s.copy()
>>> id(s)
3133387631744
>>> id(s1)
3133387631968
```

# Set Examples

```
>>> s1
{10, 20, 50, 30}
>>> s2
{40, 10, 20, 30}
>>> s1.intersection(s2)
{10, 20, 30}
>>> s1.union(s2)
{40, 10, 50, 20, 30}
>>> s1.difference(s2)
{50}
>>> s2.difference(s1)
{40}
```

# Set Examples

```
>>> s1.issubset(s2)
False
>>> s1.issuperset(s2)
False
>>> print(s1)
{10, 20, 50, 30}
>>> s1.pop()
10
>>> print(s1)
{20, 50, 30}
>>> s1.remove(50)
>>> print(s1)
{20, 30}
```

# Set Examples

```
>>> s1.remove(50)
>>> print(s1)
{20, 30}
>>> s1.update(s2)
>>> print(s1)
{20, 40, 10, 30}
```

# frozen Set

## Frozen Set

Frozen set is similar to set object but Immutable.

## Creating Frozen Set

Frozen Set in Python can be created by using the function frozenset().

# Examples

```
>>> frozenset({10,20,30,40,50})
frozenset({50, 20, 40, 10, 30})
>>> fs = frozenset({10,20,30,40,50})
>>> print(fs)
frozenset({50, 20, 40, 10, 30})
>>> print(type(fs))
<class 'frozenset'>
```

# Frozen set Methods

Frozen set Object is defined with the following pre-defined Methods

Copy, difference, intersection, isdisjoint, issubset, union.